



Chapter 3

Discrete-Time Signals in the Frequency Domain





- **The Continuous-Time Fourier Transform**
(**CTFT Review**)
- **The Discrete-Time Fourier Transform**
(**DTFT**)
 - Definition、 Properties、 Symmetry Relations
 - Theorems
 - Computation by Matlab, Unwrap function
- **Digital Processing of Continuous-Time Signals** (Sampling, Recovery, Band-pass)

How to Represent the Discrete-Time Signal?



■ **Time-domain** $\rightarrow \sum_{k=-\infty}^{\infty} a_k \delta[n - k]$

(a weighted linear combination of delayed unit sample sequences.)

■ **Frequency-domain** $\rightarrow$

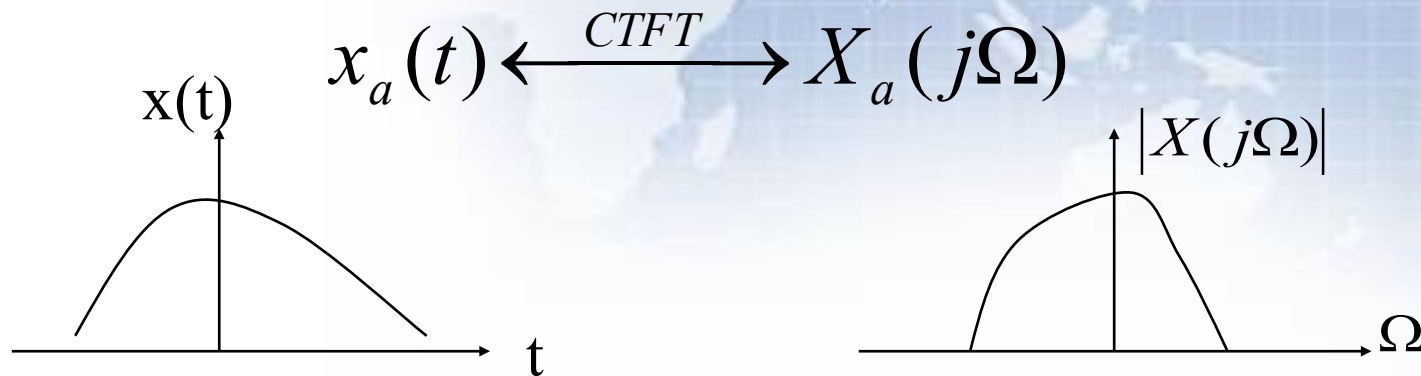
Description of a sequence in terms of complex exponential sequences of the form $\{ e^{-j\omega n} \}$.

3.1 The Continuous-Time Fourier Transform

- **CTFT – continuous in time, continuous in frequency**

$$X_a(j\Omega) = \int_{-\infty}^{\infty} x_a(t) e^{-j\Omega t} dt$$

$$x_a(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_a(j\Omega) e^{j\Omega t} d\Omega$$



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• Normally, $X_a(j\Omega)$ is a complex function.

$$X_a(j\Omega) = \text{Re}\{X_a(j\Omega)\} + j \text{Im}\{X_a(j\Omega)\}$$

• *In polar form:* $X_a(j\Omega) = |X_a(j\Omega)| e^{j\theta_a(\Omega)}$

$|X_a(j\Omega)|$ ----- *magnitude spectrum*

$\theta_a(\Omega) = \arg\{X_a(j\Omega)\}$ ----- *phase spectrum*

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• **Dirichlet conditions:** $X_a(j\Omega)$ *exists if* $x_a(t)$

- (a) Finite number of finite discontinuities and a finite number of maxima and minima in any finite interval.**
- (b) Absolutely integrable**

$$\int_{-\infty}^{\infty} |x_a(t)| dt < \infty \quad (3.3)$$

- **a milder condition of (3.3)**

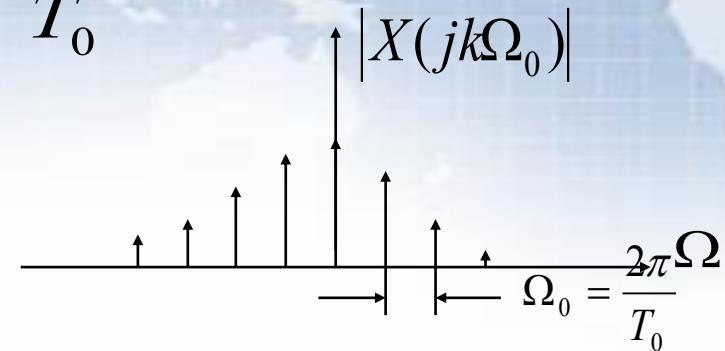
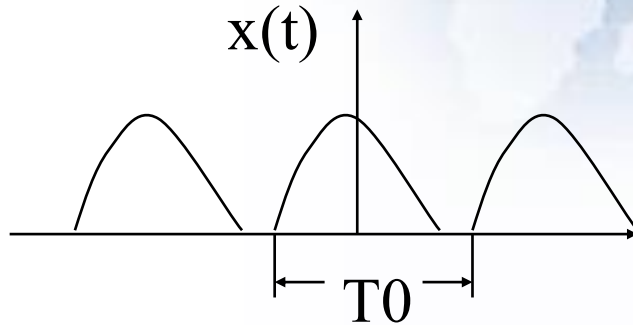
$$\int_{-\infty}^{\infty} |x_a(t)|^2 dt < \infty \quad (3.6)$$

- FS- periodicity in time, discrete in frequency

$$X(jk\Omega_0) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t) e^{-jk\Omega_0 t} dt$$

$$x(t) = \sum_{k=-\infty}^{\infty} X(jk\Omega_0) e^{jk\Omega_0 t}$$

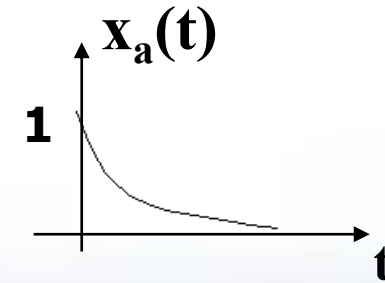
Where: $\Omega_0 = 2\pi F = \frac{2\pi}{T_0}$



Example of CTFT

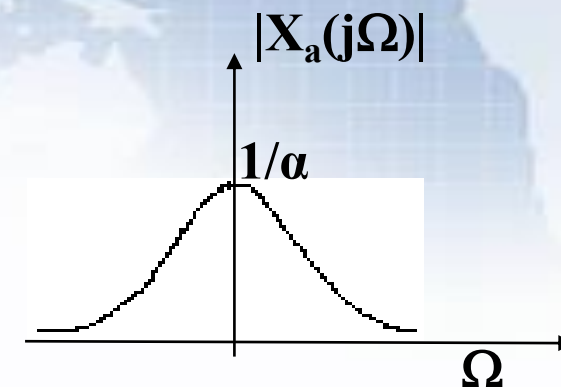


$$x_a(t) = \begin{cases} e^{-\alpha t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



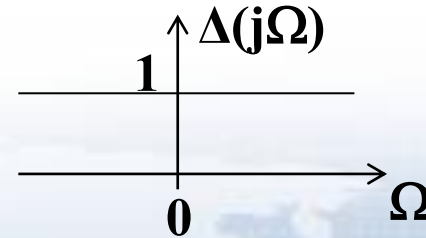
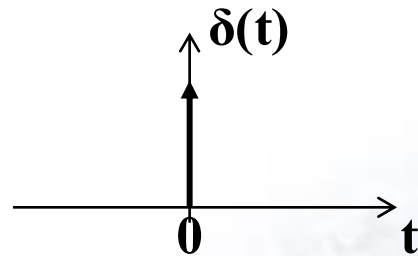
$$X_a(j\Omega) = \int_0^{\infty} e^{-\alpha t} e^{-j\Omega t} dt = \frac{1}{\alpha + j\Omega}$$

$$|X_a(j\Omega)| = \frac{1}{\sqrt{\alpha^2 + \Omega^2}}$$



$$\theta(\Omega) = -\tan^{-1} \frac{\Omega}{\alpha}$$

- Unit impulse $\delta(t)$



$$\Delta(j\Omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\Omega t} dt = 1$$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

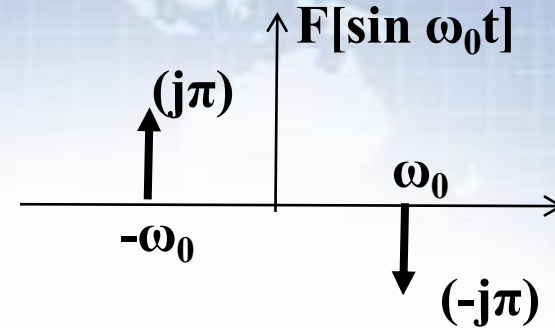
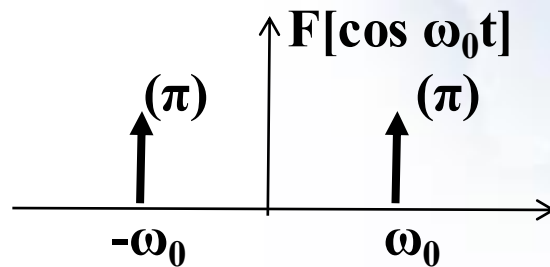
This result could be got directly based on the symmetry of Fourier Transform.

- **sin and cos**

$$F[e^{j\omega_0 t}] = \int_0^\infty e^{-j\omega_0 t} e^{-j\omega t} dt = \int_0^\infty e^{-j(\omega - \omega_0)t} dt = 2\pi\delta(\omega - \omega_0)$$

$$F[\cos\omega_0 t] = F[(e^{j\omega_0 t} + e^{-j\omega_0 t})/2] = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

$$F[\sin\omega_0 t] = F[(e^{j\omega_0 t} - e^{-j\omega_0 t})/2j] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$





- The total energy of $x_a(t)$:

$$\mathcal{E}_x = \int_{-\infty}^{\infty} |x_a(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_a(j\Omega)|^2 d\Omega$$

Parseval's Theorem

- Energy Density Spectrum:

$$S_{xx}(\Omega) = |X_a(j\Omega)|^2$$

- The energy $\mathcal{E}_{x,r}$ over frequency range $\Omega_a \leq \Omega \leq \Omega_b$ is:

$$\mathcal{E}_{x,r} = \frac{1}{2\pi} \int_{\Omega_a}^{\Omega_b} S_{xx}(\Omega) d\Omega$$

3.2 Discrete-Time Fourier Transform

3.2.1 Definition



‡ The **discrete-time Fourier transform** of a sequence $x[n]$ is defined by $X(e^{j\omega})$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

• **It's inverse transform:**

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

• **Discrete in time, periodicity in frequency**

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□ IDTFT Proof :

- The order of integration and summation can be interchanged **if** the summation inside the brackets **converges uniformly**, i.e. $X(e^{j\omega})$ exists.
- Then,

$$\begin{aligned} x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\sum_{l=-\infty}^{\infty} x[l] e^{-j\omega l} \right) e^{j\omega n} d\omega \\ &= \sum_{l=-\infty}^{\infty} x[l] \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega(n-l)} d\omega \right) = \sum_{l=-\infty}^{\infty} x[l] \frac{\sin \pi(n-l)}{\pi(n-l)} \end{aligned}$$



- Now

$$\frac{\sin \pi(n - \ell)}{\pi(n - \ell)} = \begin{cases} 1, & n = \ell \\ 0, & n \neq \ell \end{cases} = \delta[n - \ell]$$

Hence,

$$\sum_{\ell=-\infty}^{\infty} x[\ell] \frac{\sin \pi(n - \ell)}{\pi(n - \ell)} = \sum_{\ell=-\infty}^{\infty} x[\ell] \delta[n - \ell] = x[n]$$



So, DTFT is discrete in time, periodicity in frequency

- The DTFT $X(e^{j\omega})$ of a sequence $x[n]$ is a continuous function of ω
- It is also a **periodic** function of ω with a period 2π :

$$\begin{aligned} X(e^{j(\omega_o + 2\pi k)}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega_o + 2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_o n} e^{-j2\pi kn} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_o n} = X(e^{j\omega_o}) \end{aligned}$$

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- Therefore

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

represents the Fourier series representation of the periodic function.

As a result, the **Fourier coefficients $x[n]$** can be computed from $X(e^{j\omega})$ using the Fourier integral

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

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- Example

The DTFT of the unit sample sequence $\delta[n]$ is given by

$$\Delta(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = \delta[0] = 1$$



3.2.2 Basic Properties

- In general, $X(e^{j\omega})$ is a **complex function** of the real variable ω and can be written in **rectangular form** as:

$$X(e^{j\omega}) = X_{\text{re}}(e^{j\omega}) + j X_{\text{im}}(e^{j\omega})$$

Where, $X_{\text{re}}(e^{j\omega})$ and $X_{\text{im}}(e^{j\omega})$ are, respectively, the **real** and **imaginary** parts of $X(e^{j\omega})$, and are real functions of ω .

$$X_{\text{re}}(e^{j\omega}) = \frac{1}{2}\{X(e^{j\omega}) + X^*(e^{j\omega})\}$$

$$X_{\text{im}}(e^{j\omega}) = \frac{1}{2j}\{X(e^{j\omega}) - X^*(e^{j\omega})\}$$

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- $X(e^{j\omega})$ can alternately be expressed in **polar form** as

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)} \quad (3.19)$$

where

$$\theta(\omega) = \arg\{X(e^{j\omega})\} \quad (3.20)$$

- $|X(e^{j\omega})|$ is called the **magnitude function**
- $\theta(\omega)$ is called the **phase function**

✓ Both quantities are real functions of ω .

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- The relations between the rectangular and polar forms of $X(e^{j\omega})$ are given by:

$$X_{re}(e^{j\omega}) = |X(e^{j\omega})| \cos \theta(\omega)$$

$$X_{im}(e^{j\omega}) = |X(e^{j\omega})| \sin \theta(\omega)$$

$$|X(e^{j\omega})|^2 = X(e^{j\omega}) X^*(e^{j\omega}) = X_{re}^2(e^{j\omega}) + X_{im}^2(e^{j\omega})$$

$$\tan \theta(\omega) = X_{im}(e^{j\omega}) / X_{re}(e^{j\omega})$$

- For a real sequence $x[n]$, $|X(e^{j\omega})|$ and $X_{re}(e^{j\omega})$ are **even** functions of ω , whereas, $\theta(\omega)$ and $X_{im}(e^{j\omega})$ are **odd** functions of ω .



- Note: $X(e^{j\omega}) = |X(e^{j\omega})| e^{j[\theta(\omega) + 2\pi k]}$
 $= |X(e^{j\omega})| e^{j\theta(\omega)}$

for any integer k .

- The phase function $\theta(\omega)$ cannot be uniquely specified for any DTFT.
- Unless otherwise stated, we shall assume that the phase function $\theta(\omega)$ is restricted to the following range of values:

$$-\pi \leq \theta(\omega) \leq \pi$$

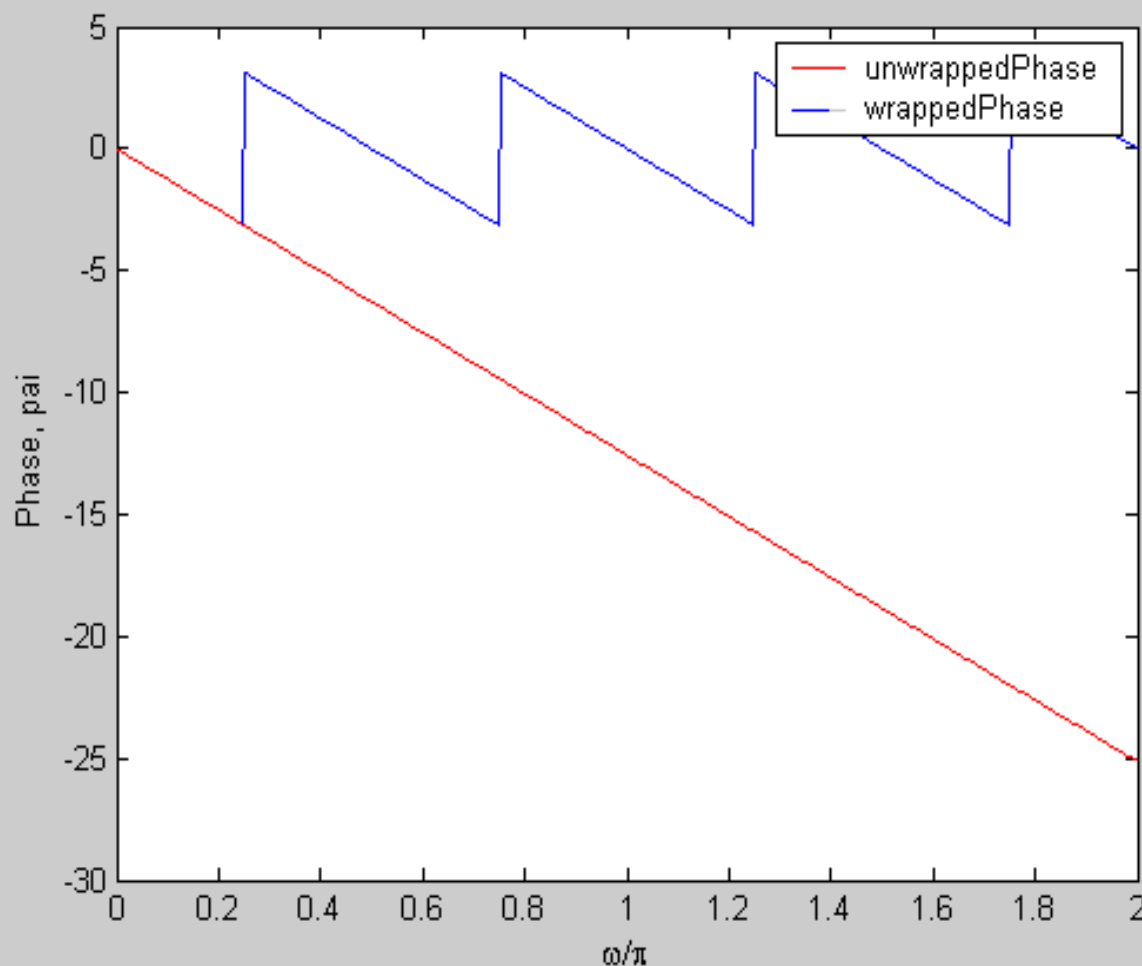
called the **principal value** (*wrapped phase*).

- The continuous function of ω is derived from the original phase function by removing the discontinuities of 2π is called **unwrapped phase**.

• *Wrapped Phase and Unwrapped Phase*



Example: $\Theta(\omega) = -4\omega$



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3.2.3 Symmetry Relations

Symmetry relations of the DTFT of a real sequence

Sequence	Discrete-Time Fourier Transform
$x[n]$	$X(e^{j\omega}) = X_{\text{re}}(e^{j\omega}) + jX_{\text{im}}(e^{j\omega})$
$x_{\text{ev}}[n]$	$X_{\text{re}}(e^{j\omega})$
$x_{\text{od}}[n]$	$jX_{\text{im}}(e^{j\omega})$

$$\begin{aligned}
 X(e^{j\omega}) &= X^*(e^{-j\omega}) \\
 X_{\text{re}}(e^{j\omega}) &= X_{\text{re}}(e^{-j\omega}) \\
 X_{\text{im}}(e^{j\omega}) &= -X_{\text{im}}(e^{-j\omega}) \\
 |X(e^{j\omega})| &= |X(e^{-j\omega})| \\
 \arg\{X(e^{j\omega})\} &= -\arg\{X(e^{-j\omega})\}
 \end{aligned}$$



Note: $x_{\text{ev}}[n]$ and $x_{\text{od}}[n]$ denote the even and odd parts of $x[n]$, respectively.

$x[n]$ is a complex sequence



Sequence	Discrete-Time Fourier Transform
$x[n]$	$X(e^{j\omega})$
$x[-n]$	$X(e^{-j\omega})$
$x^*[-n]$	$X^*(e^{j\omega})$
$\text{Re}\{x[n]\}$	$X_{\text{cs}}(e^{j\omega}) = \frac{1}{2}\{X(e^{j\omega}) + X^*(e^{-j\omega})\}$
$j\text{Im}\{x[n]\}$	$X_{\text{ca}}(e^{j\omega}) = \frac{1}{2}\{X(e^{j\omega}) - X^*(e^{-j\omega})\}$
$x_{\text{cs}}[n]$	$X_{\text{re}}(e^{j\omega})$
$x_{\text{ca}}[n]$	$jX_{\text{im}}(e^{j\omega})$

Note: $X_{\text{cs}}(e^{j\omega})$ and $X_{\text{ca}}(e^{j\omega})$ are the conjugate-symmetric and conjugate-antisymmetric parts of $X(e^{j\omega})$, respectively. Likewise, $x_{\text{cs}}[n]$ and $x_{\text{ca}}[n]$ are the conjugate-symmetric and conjugate-antisymmetric parts of $x[n]$, respectively.

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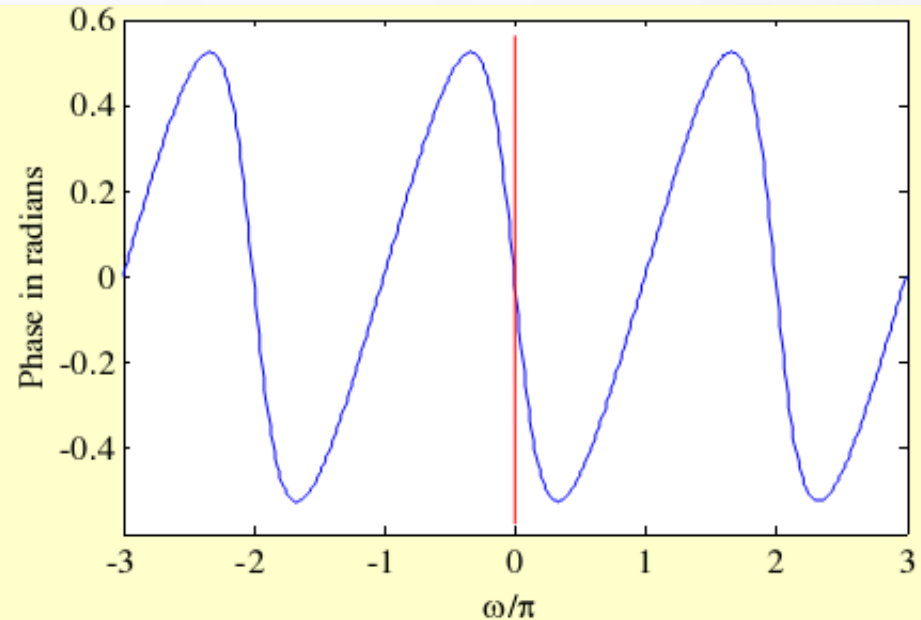
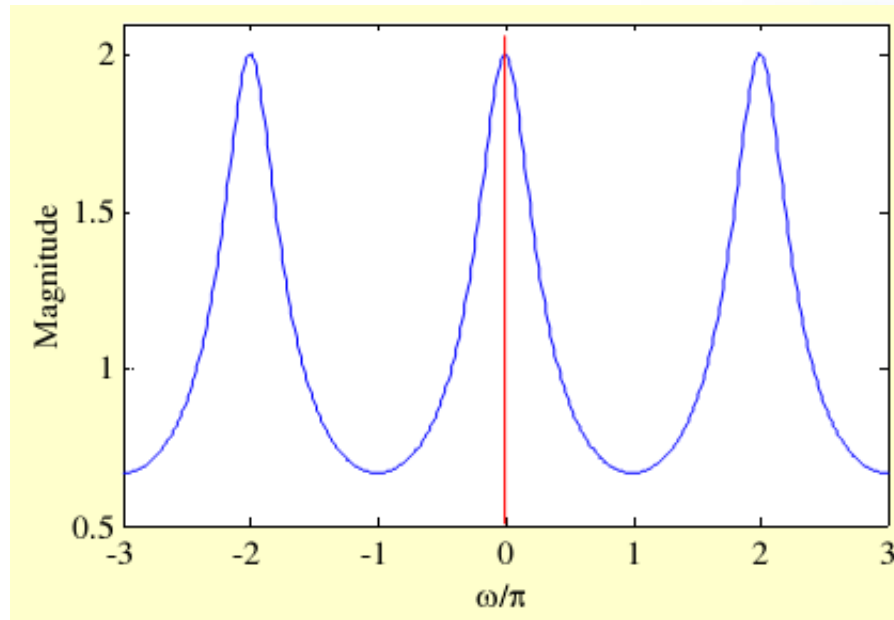
- **Example - Consider the causal sequence**

$$x[n] = \alpha^n \mu[n], \quad |\alpha| < 1$$

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} \alpha^n \mu[n] e^{-j\omega n} = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n = \frac{1}{1 - \alpha e^{-j\omega}} \quad \text{as } |\alpha e^{-j\omega}| = |\alpha| < 1 \end{aligned}$$

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 - 2\alpha \cos \omega + \alpha^2}} \quad \theta(\omega) = \tan^{-1} \left(\frac{-\alpha \sin \omega}{1 - \alpha \cos \omega} \right)$$

- The magnitude and phase of the $X(e^{j\omega}) = 1/(1 - 0.5e^{-j\omega})$ are shown below:



$$|X(e^{j\omega})| = |X(e^{-j\omega})|$$

$$\theta(\omega) = -\theta(-\omega)$$

3.2.4 Convergence Condition



- If we denote:
$$X_K(e^{j\omega}) = \sum_{n=-K}^K x[n]e^{-j\omega n}$$

- The **uniform convergence** of $X(e^{j\omega})$:

$$\lim_{K \rightarrow \infty} |X(e^{j\omega}) - X_K(e^{j\omega})| = 0$$

- The **mean-square convergence** of $X(e^{j\omega})$:

$$\lim_{k \rightarrow \infty} \int_{-\pi}^{\pi} |X(e^{j\omega}) - X_K(e^{j\omega})|^2 d\omega = 0$$

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- Now, if $x[n]$ is an *absolutely summable* sequence, i.e., if:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

- Then

$$\left| X(e^{j\omega}) \right| = \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

for all values of ω .

Thus, the absolute summability of $x[n]$ is a *sufficient condition* for the existence of the DTFT $X(e^{j\omega})$.

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- Since

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 \leq \left(\sum_{n=-\infty}^{\infty} |x[n]| \right)^2$$

an absolutely summable sequence has always a finite energy.

However, a finite-energy sequence is not necessarily absolutely summable.

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*Uniform
convergence*

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \xrightarrow{\text{sufficient}} X(e^{j\omega}) \text{ exist}$$

Absolutely summable sequence.

$$\sum_{n=-\infty}^{\infty} (x[n])^2 < \infty \xrightarrow{\text{sufficient}} X(e^{j\omega}) \text{ exist}$$

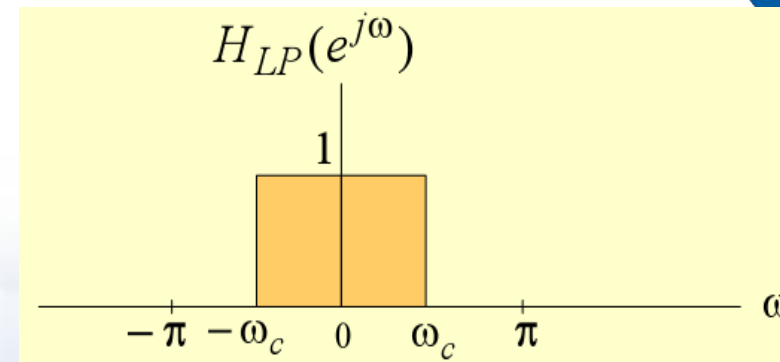
Finite-energy sequence

*Mean-square
convergence*

• Example 3.8



$$H_{LP}(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| \leq \omega_c \\ 0, & \omega_c \leq |\omega| \leq \pi \end{cases}$$



$$h_{LP}[n] = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\sin \omega_c n}{\pi n}, & n \neq 0 \end{cases}$$

$$\sum_{n=-\infty}^{\infty} h_{LP}[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \frac{\sin \omega_c n}{\pi n} e^{-j\omega n}$$

does **not converge uniformly** for all values of ω , but converge in the **mean-square** sense.

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- As a result

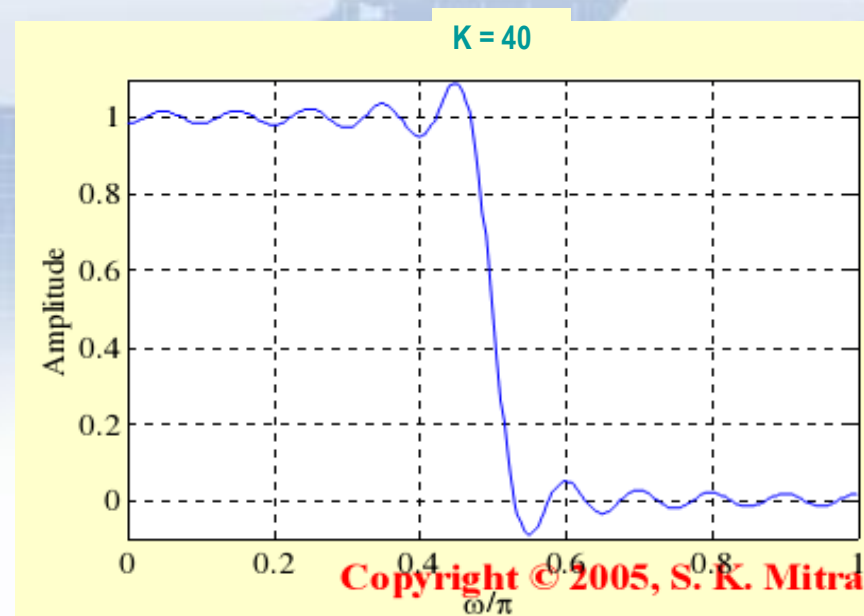
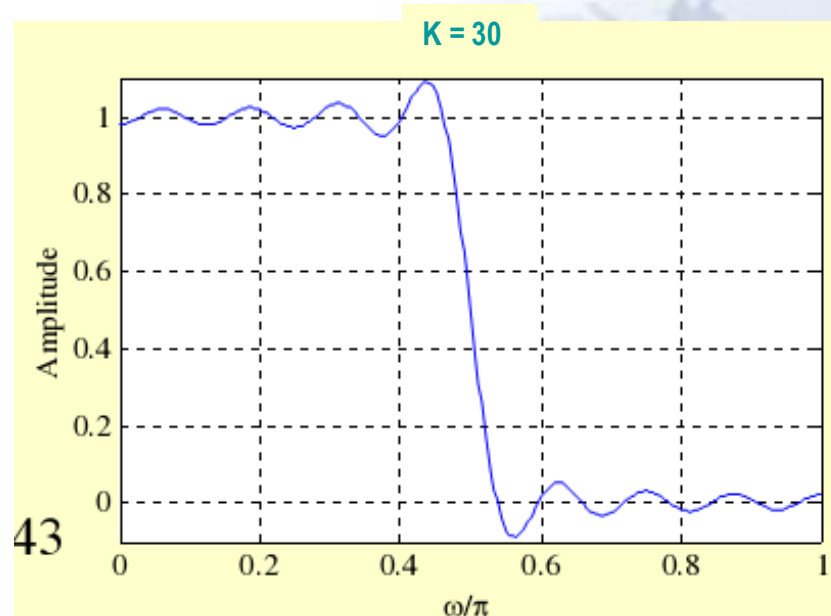
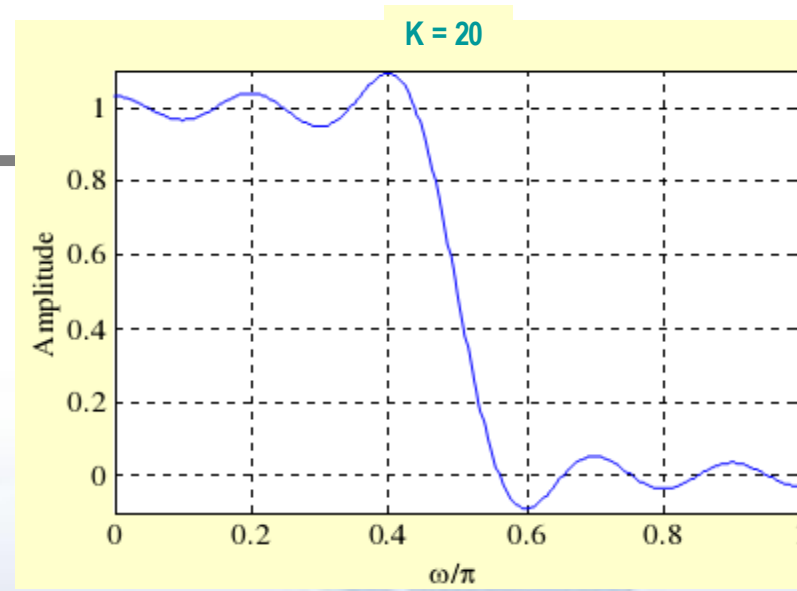
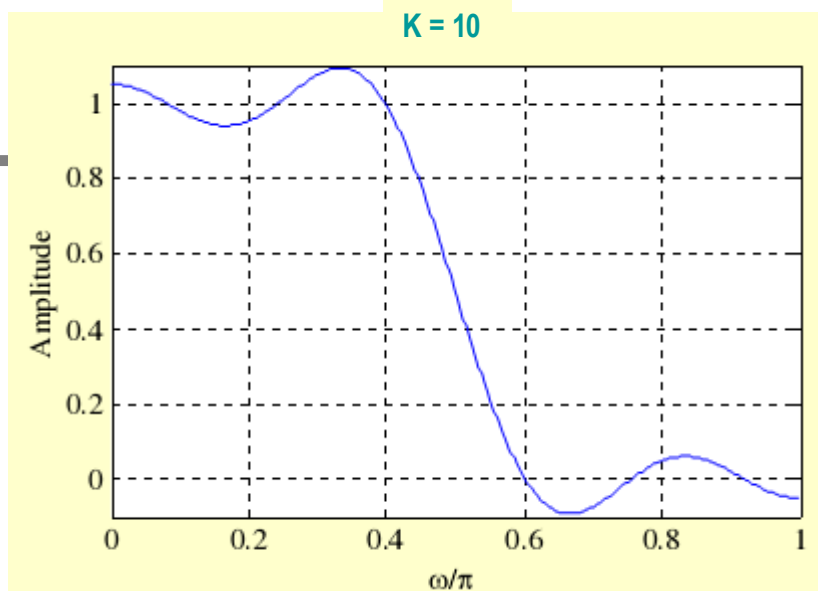
$$\sum_{n=-K}^K h_{LP}[n]e^{-j\omega n} = \sum_{n=-K}^K \frac{\sin \omega_c n}{\pi n} e^{-j\omega n}$$

Does not **uniformly converge** to $H_{LP}(e^{j\omega})$ for all values of ω , but converges to $H_{LP}(e^{j\omega})$ **in the mean-square sense**

- The mean-square convergence property of the sequence $h_{LP}[n]$ can be further illustrated by examining the plot of the function

$$H_{LP,K}(e^{j\omega}) = \sum_{n=-K}^K \frac{\sin \omega_c n}{\pi n} e^{-j\omega n}$$

For various values of K as shown in next slide



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- As can be seen from these plots, independent of the value of K there are ripples in the plot of $H_{LP,K}(e^{j\omega})$ around both sides of the point $\omega = \omega_c$
- The number of ripples increases as K increases with the height of the largest ripple remaining the same for all values of K



- As K goes to infinity, the condition

$$\lim_{K \rightarrow \infty} \int_{-\pi}^{\pi} |H_{LP}(e^{j\omega}) - H_{LP,K}(e^{j\omega})|^2 d\omega = 0$$

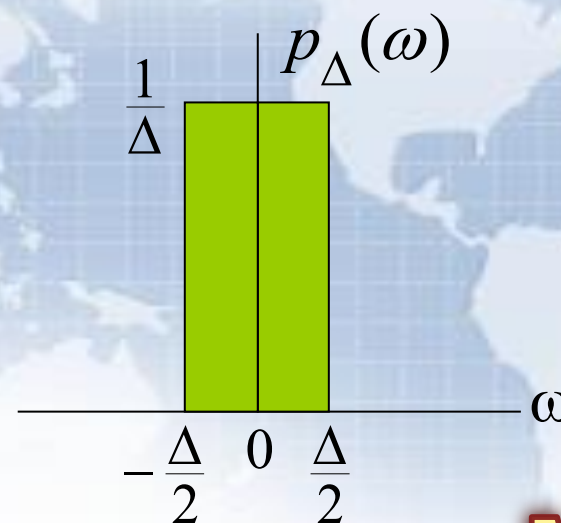
holds indicating the convergence of $H_{LP,K}(e^{j\omega})$ approximation $H_{LP}(e^{j\omega})$ in the mean-square sense at a point of discontinuity is known as the Gibbs phenomenon.



- The DTFT can also be defined for a certain class of sequences which are **neither absolutely summable nor square summable**.
- Examples of such sequences are the unit step sequence $\mu[n]$, the sinusoidal sequence $\cos(\omega_0 n + \varphi)$ and the exponential sequence $A\alpha^n$
- For this type of sequences, a DTFT representation is possible using the Dirac Delta function $\delta(\omega)$

- A Dirac Delta function $\delta(\omega)$ is a function of ω with infinite height, zero width, and unit area
- It is the limiting form of a unit area pulse function $p_{\Delta}(\omega)$ as Δ goes to zero satisfying:

$$\lim_{\Delta \rightarrow 0} \int_{-\infty}^{\infty} p_{\Delta}(\omega) d\omega = \int_{-\infty}^{\infty} \delta(\omega) d\omega$$



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- **Example 3.9 - Consider the complex exponential sequence**

$$x[n] = e^{j\omega_o n}$$

Its DTFT is given by

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_o + 2\pi k)$$

where $\delta(\omega)$ is an impulse function of ω and

$$-\pi \leq \omega_o \leq \pi$$

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- **The function**

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_o + 2\pi k)$$

is a periodic function of ω with a period 2π and is called a **periodic impulse train** or **impulse train**.

- To verify that $X(e^{j\omega})$ given above is indeed the DTFT of $x[n]=e^{j\omega_0 n}$ we compute the inverse DTFT of $X(e^{j\omega})$

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- Thus

$$\begin{aligned}x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_o + 2\pi k) e^{j\omega n} d\omega \\&= \int_{-\pi}^{\pi} \delta(\omega - \omega_o) e^{j\omega n} d\omega = e^{j\omega_o n}\end{aligned}$$

where we have used the sampling property of the impulse function $\delta(\omega)$

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Commonly Used DTFT Pairs

(Table 3.3 Page 105)



$$\delta[n] \leftrightarrow 1$$

$$1 \leftrightarrow \sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$$

$$\mu[n] \leftrightarrow \frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$$

$$e^{j\omega_0 n} \leftrightarrow \sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$$

$$\alpha^n \mu[n] (|\alpha| < 1) \leftrightarrow \frac{1}{1 - \alpha e^{-j\omega}}$$

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3.2.5 Strength of a DTFT

- The strength of a DTFT is given by its norm.
- The L_p -norm of $X(e^{j\omega})$ is defined by:

$$\|X\|_p = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^p d\omega \right)^{1/p}$$

p is a positive integer.

- In practice, the value of p is typically 1 or 2 or ∞ .

■ M-file *filternorm* can be used to determine norms.

3.3 DTFT Theorems (Table 3.4)



$$g[n] \leftrightarrow G(e^{j\omega}) \quad h[n] \leftrightarrow H(e^{j\omega})$$

Linearity $\alpha g[n] + \beta h[n] \leftrightarrow \alpha G(e^{j\omega}) + \beta H(e^{j\omega})$

Time-shifting $g[n - n_0] \leftrightarrow e^{-j\omega n_0} G(e^{j\omega})$

Frequency-shifting $e^{j\omega_0 n} g[n] \leftrightarrow G(e^{j(\omega - \omega_0)})$

Differentiation-in-frequency $ng[n] \leftrightarrow j \frac{dG(e^{j\omega})}{d\omega}$

Convolution $g[n] \otimes h[n] \leftrightarrow G(e^{j\omega}) H(e^{j\omega})$

Modulation $g[n] h[n] \leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} G(e^{j\theta}) H(e^{j(\omega - \theta)}) d\theta$

Parseval's relation $\sum_{n=-\infty}^{\infty} g[n] h^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} G(e^{j\omega}) H^*(e^{j\omega}) d\omega$

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Application of Above Properties

- **Compute DTFT of sequences conveniently**

Ex. 3.10, 3.11, 3.12

- **Compute linear convolution:**

$$Q \quad g[n] \otimes h[n] \leftrightarrow G(e^{j\omega})H(e^{j\omega})$$

$$\begin{array}{l} g[n] \leftrightarrow G(e^{j\omega}) \quad \text{---} \rightarrow \\ h[n] \leftrightarrow H(e^{j\omega}) \quad \text{---} \rightarrow \end{array} \quad G(e^{j\omega})H(e^{j\omega}) \xrightarrow{\text{IDTFT}} g[n] \otimes h[n]$$

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- Example 3.13

Determine the DTFT $Y(e^{j\omega})$ of $y[n]=(n+1)\alpha^n\mu[n]$,
 $|\alpha|<1$

Let $x[n]=\alpha^n\mu[n]$, $|\alpha|<1$

We can therefore write

$$y[n]=nx[n] + x[n]$$

- The DTFT of $x[n]$ is given by

$$X(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}$$

- Using the differentiation property of the DTFT, we observe that the DTFT of $n x[n]$ is given by

$$j \frac{dX(e^{j\omega})}{d\omega} = j \frac{d}{d\omega} \left(\frac{1}{1 - \alpha e^{-j\omega}} \right) = \frac{\alpha e^{-j\omega}}{(1 - \alpha e^{-j\omega})^2}$$

- Next using the linearity property of the DTFT, we have:

$$Y(e^{j\omega}) = \frac{\alpha e^{-j\omega}}{(1 - \alpha e^{-j\omega})^2} + \frac{1}{1 - \alpha e^{-j\omega}} = \frac{1}{(1 - \alpha e^{-j\omega})^2}$$

• Example 3.14 Convolution Sum Computation Using DTFT



$$x[n] = \alpha^n \mu[n], |\alpha| < 1; h[n] = \beta^n \mu[n], |\beta| < 1$$

Compute $y[n] = x[n] * h[n]$

$$\alpha^n \mu[n] \xleftrightarrow{\text{DTFT}} \frac{1}{1 - \alpha e^{-j\omega}}, \quad \beta^n \mu[n] \xleftrightarrow{\text{DTFT}} \frac{1}{1 - \beta e^{-j\omega}}$$

$$\begin{aligned} Y(e^{j\omega}) &= X(e^{j\omega})H(e^{j\omega}) = \frac{1}{(1 - \alpha e^{-j\omega})(1 - \beta e^{-j\omega})} \\ &= \frac{\alpha/(\alpha - \beta)}{1 - \alpha e^{-j\omega}} - \frac{\beta/(\alpha - \beta)}{1 - \beta e^{-j\omega}} \end{aligned}$$

$$\therefore y[n] = \frac{\alpha}{\alpha - \beta} \alpha^n \mu[n] - \frac{\beta}{\alpha - \beta} \beta^n \mu[n] = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} \mu[n]$$

MOOC

3.4 Energy Density Spectrum of a Discrete-Time Sequence



- The total energy of a finite-energy sequence $g[n]$ is given by

$$\mathcal{E}_g = \sum_{n=-\infty}^{\infty} |g[n]|^2$$

- From Parseval's relation we observe that

$$\mathcal{E}_g = \sum_{n=-\infty}^{\infty} |g[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |G(e^{j\omega})|^2 d\omega$$

MOOC



- **The energy density spectrum:**

$$S_{gg}(\omega) = |G(e^{j\omega})|^2$$

- The area under this curve in the range $\pi \leq \omega \leq \pi$ divided by 2π is the energy of the sequence.

MOOC



- **Example 3.15** Compute the energy of the sequence

$$h_{LP}[n] = \sin \omega_c n / \pi n, \quad -\infty < n < \infty$$

$$\sum_{n=-\infty}^{\infty} |h_{LP}[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H_{LP}(e^{j\omega})|^2 d\omega$$

Where, $H_{LP}(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$

$$\therefore \sum_{n=-\infty}^{\infty} |h_{LP}[n]|^2 = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega = \frac{\omega_c}{\pi} < \infty$$

• **Hence, $h_{LP}[n]$ is a finite-energy lowpass sequence.**



• Example 3.16

Compute the energy of the exponential sequence

$$\alpha^n \mu[n] \xleftrightarrow{DTFT} \frac{1}{1 - \alpha e^{-j\omega}}$$

From Parseval's relation

$$e_x = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \frac{1}{1 - \alpha e^{-j\omega}} \right|^2 d\omega = \sum_{n=0}^{\infty} |\alpha|^{2n} = \frac{1}{1 - |\alpha|^2}$$

When $\alpha = 0.5$, $e_x = 1.3333$

3.1.3 Band-Limited Continuous-Time Signals



- An ideal band-limited signal has a spectrum that is zero outside a frequency range

$0 < \Omega_a \leq |\Omega| \leq \Omega_b < \infty$, that is

$$X_a(j\Omega) = \begin{cases} 0, & 0 \leq |\Omega| \leq \Omega_a \\ 0, & \Omega_b \leq |\Omega| \leq \infty \end{cases}$$

Band-limited signals are classified according to the frequency range where most of the signal's **energy is concentrated.**



- A **lowpass** continuous-time signal has a spectrum occupying the frequency range $0 < |\Omega| \leq \Omega_p < \infty$, where Ω_p is called the **bandwidth** of the signal.
- A **highpass** continuous-time signal has a spectrum occupying the frequency range $0 < \Omega_p \leq |\Omega| < \infty$ and has a bandwidth from Ω_p to ∞ .
- A **bandpass** continuous-time signal has a spectrum occupying the frequency range $0 < \Omega_L \leq |\Omega| \leq \Omega_H < \infty$ and has a **bandwidth** of $\Omega_H \sim \Omega_L$



- A precise definition of the **bandwidth** depends on applications.
- As can be seen from Figure 3.2(a), the signal in example 3.1 is a lowpass signal. It can be shown that 80% energy of this signal is contained in the frequency range $0 \leq |\Omega| \leq 0.4898\pi$
- Hence, we can define the **80% bandwidth** to be 0.4898π

3.5 *Band-Limited Discrete-Time Signals*



- Since the spectrum of a discrete-time signal is a periodic function of ω with a period 2π , a full-band signal has a spectrum occupying the frequency range $-\pi \leq \omega \leq \pi$.
- A **band-limited** discrete-time signal has a spectrum that is limited to a portion of the frequency range $-\pi \leq \omega \leq \pi$.



- **An ideal band-limited signal is defined as:**

$$X(e^{j\omega}) = \begin{cases} 0, & 0 \leq |\omega| \leq \omega_a \\ 0, & \omega_b \leq |\omega| \leq \pi \end{cases}$$

- **An ideal band-limited discrete-time signal cannot be generated in practice.**
- **In practice, we can make the signal energy be very small outside the specified frequency range to get band-limited signal.**



- A lowpass discrete-time real signal has a spectrum occupying the frequency range $0 < |\omega| \leq \omega_p < \pi$ and has a bandwidth of ω_p ;
- A highpass discrete-time real signal has a spectrum occupying the frequency range $0 < \omega_p \leq |\omega| < \pi$ and has a bandwidth of $\pi - \omega_p$;
- A bandpass discrete-time real signal has a spectrum occupying the frequency range $0 < \omega_L \leq |\omega| \leq \omega_H < \pi$ and has a bandwidth of $\omega_H - \omega_L$;

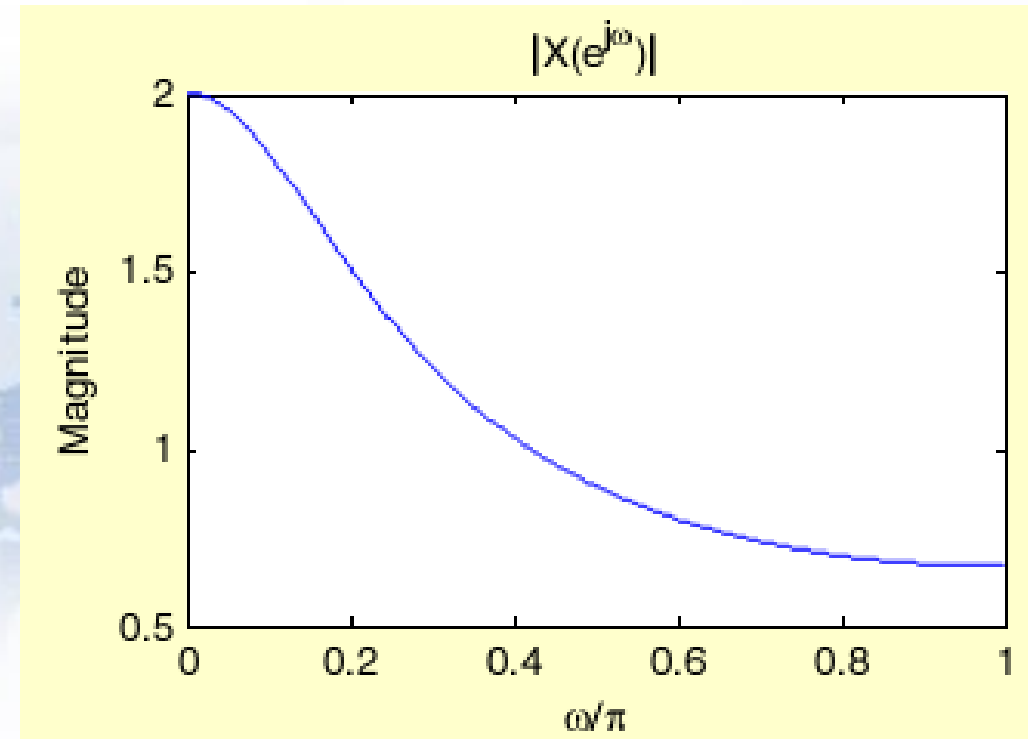
- Example:

Consider the sequence

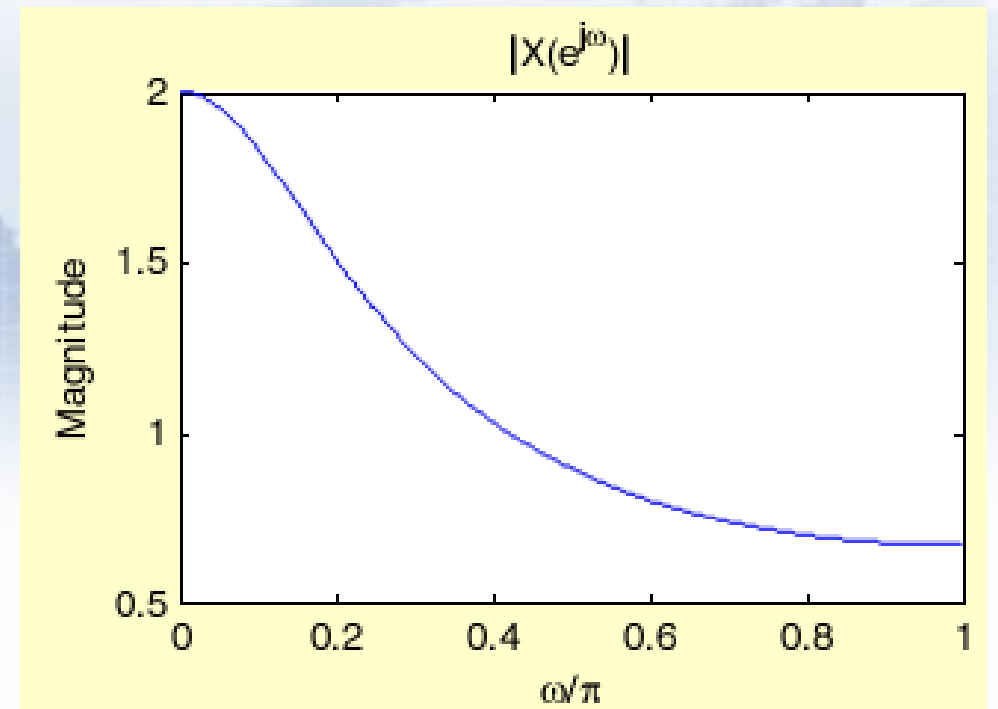
$$x[n] = (0.5)^n \mu[n]$$

- Its DTFT is given below on the left along with its magnitude spectrum shown below on the right

$$X(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$



- It can be shown that 80% of the energy of this lowpass signal is contained in the frequency range $0 \leq |\omega| \leq 0.5081\pi$
- Hence, we can define the 80% bandwidth to be 0.5081π



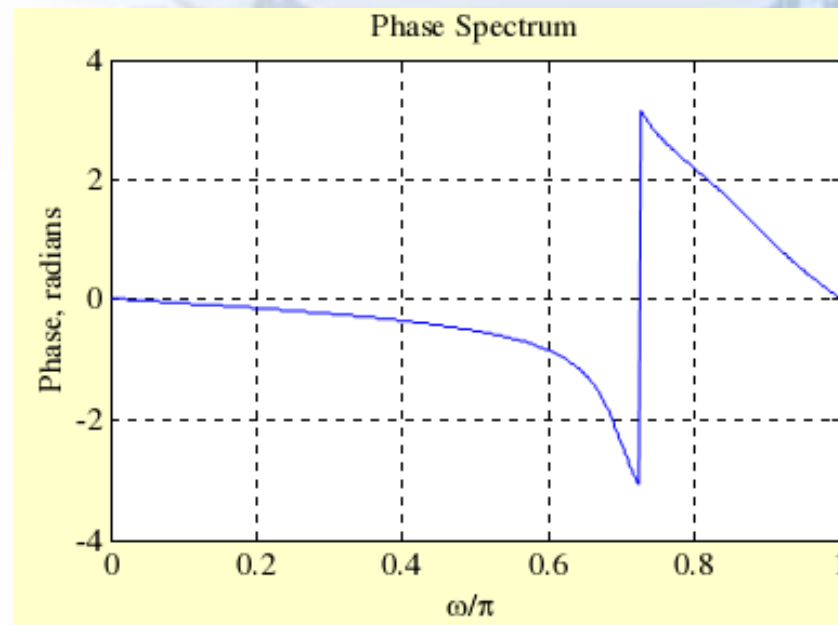


3.7 *The Unwrapped Phase Function*

- In numerical computation, when the computed phase function is outside the range $[-\pi, \pi]$, the phase is computed modulo 2π , to bring the computed value to this range.
- Thus, the phase functions of some sequences exhibit discontinuities of 2π radians in the plot.

- For example, there is a discontinuity of 2π at $\omega=0.72$ in the phase response below:

$$X(e^{j\omega}) = \frac{0.008 - 0.033e^{-j\omega} + 0.05e^{-j2\omega} - 0.033e^{-j3\omega} + 0.008e^{-j4\omega}}{1 + 2.37e^{-j\omega} + 2.7e^{-j2\omega} + 1.6e^{-j3\omega} + 0.41e^{-j4\omega}}$$





- In such cases, often an alternate type of phase function that is continuous function of ω is derived from the original phase function by removing the discontinuities of 2π .
- Process of discontinuity removal is called **unwrapping** the phase.
- The unwrapped phase function will be denoted as $\theta_c(\omega)$.
- In Matlab, this can be done by using ***unwrap***.



■ The **conditions** under which the phase function will be continuous function of ω .

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)}$$

$$\text{Where, } \theta(\omega) = \arg\{X(e^{j\omega})\}$$

$$\text{so, } \ln X(e^{j\omega}) = \ln |X(e^{j\omega})| + j\theta(\omega)$$

$$\text{then, } \frac{d \ln X(e^{j\omega})}{d\omega} = \frac{d \ln |X(e^{j\omega})|}{d\omega} + j \frac{d\theta(\omega)}{d\omega}$$

• $\theta(\omega)$ can be defined unequivocally by its derivative $\frac{d\theta(\omega)}{d\omega}$



- **Unwrapped phase:**

$$\theta(\omega) = \int_0^{\omega} \left[\frac{d\theta(\eta)}{d\eta} \right] d\eta$$

With constraint: $\theta(0) = 0$

$\theta(\omega)$ will be continuous function of ω .

- **Moreover, if** $\frac{1}{\pi} \int_0^{2\pi} \left[\frac{d\theta(\eta)}{d\eta} \right] d\eta = 0$

$\theta(\omega)$ will be odd function of ω .



3.6 DTFT Computation Using MATLAB

- The function *freqz*, *abs*, *angle*, *unwrap* can be used to compute the values of the DTFT of a sequence, described as a rational function in the form of

$$X(e^{j\omega}) = \frac{p_0 + p_1 e^{-j\omega} + \dots + p_M e^{-j\omega M}}{d_0 + d_1 e^{-j\omega} + \dots + d_N e^{-j\omega N}}$$

at a prescribed set of discrete frequency points $\omega = \omega_\lambda$



- For example, the statement

$H = \text{freqz}(\text{num}, \text{den}, w)$

returns the frequency response values as a vector **H** of a DTFT defined in terms of the vectors **num** and **den** containing the coefficients $\{p_i\}$ and $\{d_i\}$, respectively at a prescribed set of frequencies between 0 and 2π given by the vector ω .

- There are several other forms of the function **freqz**
- With the functions of **real**, **imag**, **abs**, **angle**, it computes the real and imaginary parts, and the magnitude and phase of the DTFT.



- Example
- Plots of the real and imaginary parts, and the magnitude and phase of the DTFT

$$X(e^{j\omega}) = \frac{0.008 - 0.033e^{-j\omega} + 0.05e^{-j2\omega} - 0.033e^{-j3\omega} + 0.008e^{-j4\omega}}{1 + 2.37e^{-j\omega} + 2.7e^{-j2\omega} + 1.6e^{-j3\omega} + 0.41e^{-j4\omega}}$$



% Program 3_1: Discrete-Time Fourier Transform Computation

% Generate the filter coefficients

```
num = [0.008 -0.033 0.05 -0.033 0.008];
```

```
den = [1 2.37 2.7 1.6 0.41];
```

```
w = 0:pi/255:pi;
```

```
h = freqz(num, den, w); % Compute the frequency responses
```

```
subplot(2,2,1)
```

```
plot(w/pi,real(h));grid
```

```
title('Real part'); xlabel('\omega/\pi'); ylabel('Amplitude')
```

```
subplot(2,2,2)
```

```
plot(w/pi,imag(h));grid
```

```
title('Imaginary part'); xlabel('\omega/\pi'); ylabel('Amplitude')
```

```
subplot(2,2,3)
```

```
plot(w/pi,abs(h));grid
```

```
title('Magnitude Spectrum'); xlabel('\omega/\pi'); ylabel('Magnitude')
```

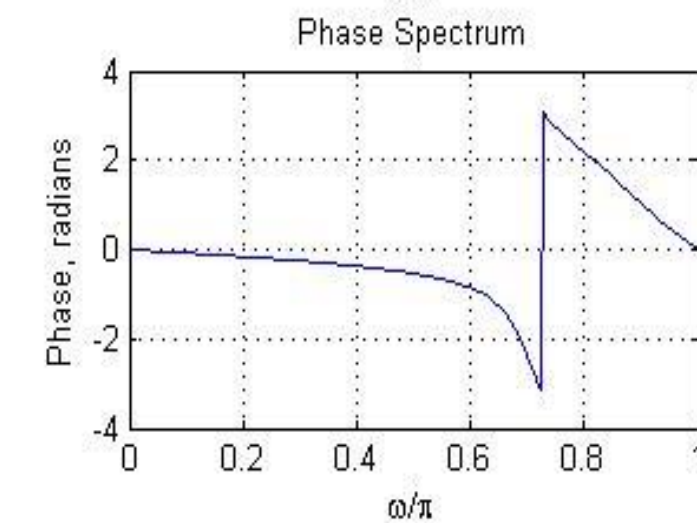
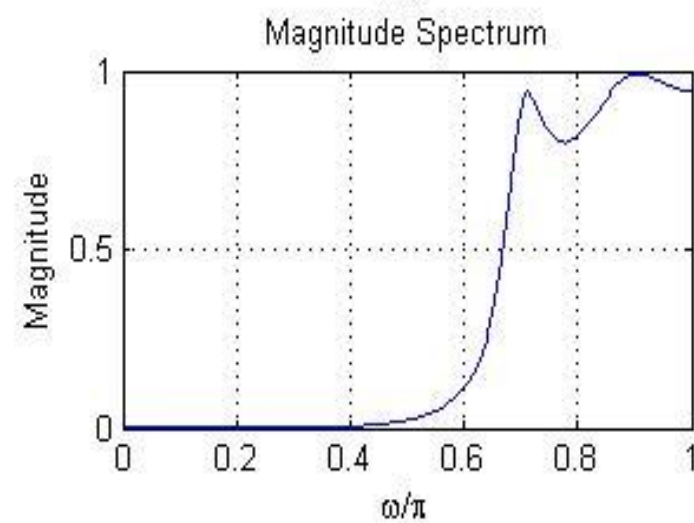
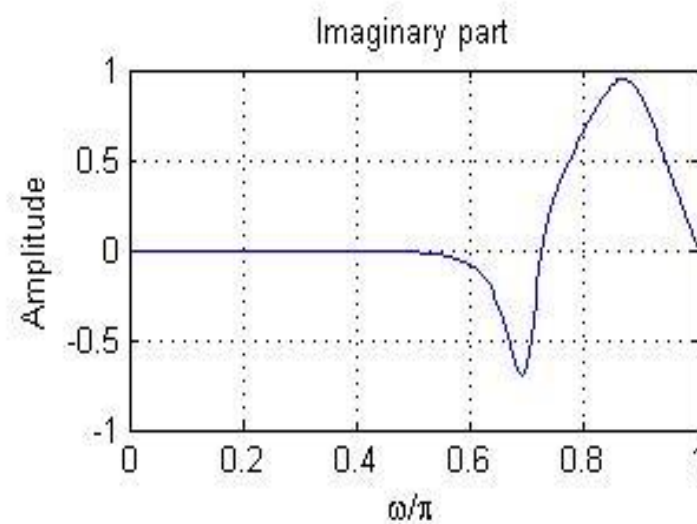
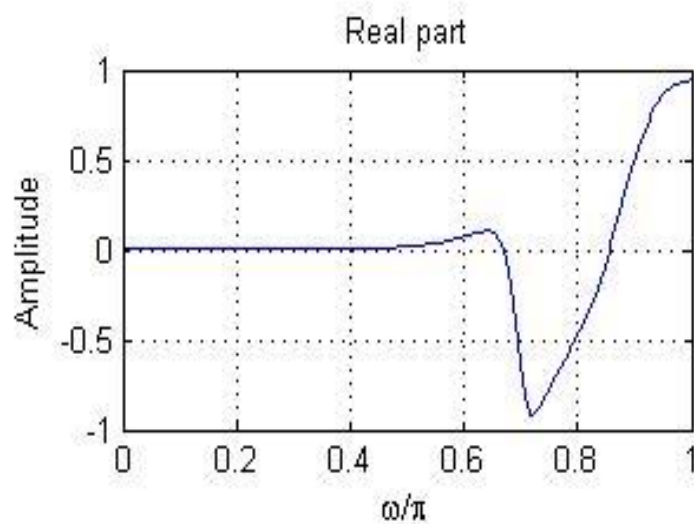
```
subplot(2,2,4)
```

```
plot(w/pi,angle(h));grid
```

```
title('Phase Spectrum'); xlabel('\omega/\pi'); ylabel('Phase, radians')
```

```
figure, plot(w/pi,unwrap(angle(h)));
```

```
title('unwrapped Phase Spectrum'); xlabel('\omega/\pi'); ylabel('Phase, radians')
```



- The phase spectrum displays a discontinuity of 2π at $\omega = 0.72$
- This discontinuity can be removed using the function *unwrap* as indicated below:

